BLEEP EDUCATION

ARTIFICIAL INTELLIGENCE INTERNSHIP

JUNE BATCH

INTERNSHIP PROJECT REPORT

EMPLOYEE ATTRITION & WORKFORCE ANALYTICS

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Exploratory Data Analysis (EDA)

Submitted by

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Employee attrition is an important workforce analytics problem because employee characteristics, work conditions, satisfaction, compensation, and organizational factors can be examined to understand patterns associated with employees leaving an organization. This project presents an Exploratory Data Analysis (EDA) of an employee workforce dataset containing 1,470 records and 35 initial features.

The analysis begins with dataset loading and structural inspection, followed by numerical and categorical summaries, missing-value analysis, unique-value analysis, visualization of the attrition distribution, and exploration of relationships involving overtime, job satisfaction, department, age, monthly income, distance from home, and years at the company. A correlation heatmap is used to examine linear relationships among numerical variables, while boxplots and scatterplots provide additional views of distributions and relationships.

The notebook shows that attrition is imbalanced, with 83.88% of records corresponding to employees who did not leave and 16.12% corresponding to employees who left. Overtime, job satisfaction, department, age, monthly income, distance from home, and years at the company are examined as descriptive factors associated with attrition. Several strong numerical correlations are also visible among job level, income, total working

years, and tenure-related variables. Constant features were removed before the outlier analysis. The findings are descriptive and do not establish causation.

The project demonstrates a structured EDA workflow and provides a foundation for deeper statistical analysis and, where appropriate, subsequent predictive modeling.

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1. INTRODUCTION

1.1 Background

Exploratory Data Analysis is a foundational stage of data analytics in which the structure, quality, distributions, and relationships present in a dataset are examined before advanced statistical or machine-learning work. In a workforce context, EDA can be used to describe employee characteristics and investigate how workforce variables are distributed across different attrition outcomes.

1.2 Problem Context

The project focuses on employee attrition: the observed outcome indicating whether an employee left the organization. The dataset contains demographic, job-related, compensation, satisfaction, work-life, and tenure-related variables. The analysis therefore examines both the overall workforce structure and selected relationships with the Attrition variable.

1.3 Purpose of the Analysis

The purpose of the project is to understand the employee dataset through systematic exploratory analysis, identify descriptive patterns related to attrition, visualize important relationships, and prepare a reliable analytical foundation for subsequent work.

1.4 Project Objectives

- Understand the dimensions, structure, data types, and feature inventory of the dataset.
- Summarize numerical and categorical variables using descriptive statistics.
- Check for missing values and identify features with no useful variation.
- Explore the distribution of employee attrition and selected workforce characteristics.
- Analyze descriptive relationships involving overtime, job satisfaction, department, age, income, distance from home, and tenure.
- Examine numerical correlations using a correlation heatmap.
- Use boxplots and scatterplots to investigate distributions and relationships.
- Document key findings without interpreting descriptive associations as causal effects.

1.5 Scope of the Project

The scope is limited to the completed EDA notebook. It includes data understanding, descriptive analysis, visualization, correlation analysis, feature removal of constant/identifier fields, and outlier inspection. Predictive model training and model-performance evaluation are outside the scope of this final EDA report.

2. DATASET OVERVIEW

2.1 Dataset Description

The notebook loads the file WA_Fn-UseC_-HR-Employee-Attrition.csv into a Pandas DataFrame. The dataset contains employee-level records with demographic, organizational, compensation, satisfaction, and employment-history variables, together with the Attrition outcome.

2.2 Dataset Source

The notebook documents the dataset by its local CSV filename and loading path. It does not record an external dataset URL or publisher. Accordingly, this report identifies the source as the dataset file used by the notebook rather than attributing it to an external source not documented in the project material.

2.3 Number of Records and Features

Property	Value
Initial records	1,470
Initial features	35
Numerical features	26
Categorical/string features	9
Records after feature removal	1,470
Features after feature removal	31

2.4 Numerical and Categorical Features

The notebook's df.info() output identifies 26 numerical features stored as int64 and 9 categorical/string features. The categorical fields include Attrition, BusinessTravel, Department, EducationField, Gender, JobRole, MaritalStatus, Over18, and OverTime.

2.5 Target Variable

Attrition is the target variable. Its two observed categories are No and Yes. The notebook records 1,233 employees with Attrition = No and 237 employees with Attrition = Yes.

Attrition category	Count	Percentage
No	1,233	83.877551%
Yes	237	16.122449%
Total	1,470	100%

2.6 Important Features / Data Dictionary

The following feature inventory is taken directly from the notebook. The descriptions are limited to the feature names and their analytical role demonstrated in the notebook.

Feature	Type	Use / role evidenced in notebook
Age	Numerical	Used in distribution, boxplot, and Age–MonthlyIncome scatter analysis.
Attrition	Categorical	Target outcome: No / Yes.
BusinessTravel	Categorical	Categorical workforce attribute inspected in summaries.
DailyRate	Numerical	Included in numerical summary and correlation analysis.
Department	Categorical	Compared with attrition and shown in department distribution.
DistanceFromHome	Numerical	Compared with attrition and included in outlier analysis.
Education	Numerical	Included in numerical summary and correlation analysis.
EducationField	Categorical	Included in categorical summary.
EmployeeCount	Numerical	Constant feature; removed.
EmployeeNumber	Numerical	Identifier-like field; removed.
EnvironmentSatisfaction	Numerical	Included in numerical summary/correlation

Feature	Type	Use / role evidenced in notebook
		analysis.
Gender	Categorical	Shown in gender distribution.
HourlyRate	Numerical	Included in numerical summary/correlation analysis.
JobInvolvement	Numerical	Included in numerical summary/correlation analysis.
JobLevel	Numerical	Included in numerical summary/correlation analysis.
JobRole	Categorical	Included in categorical summary.
JobSatisfaction	Numerical	Compared with attrition using countplot.
MaritalStatus	Categorical	Included in categorical summary.
MonthlyIncome	Numerical	Compared with attrition and Age/YearsAtCompany in plots.
MonthlyRate	Numerical	Included in numerical summary/correlation analysis.
NumCompaniesWorked	Numerical	Included in numerical summary/correlation analysis.
Over18	Categorical	Constant feature; removed.
OverTime	Categorical	Distribution shown and attrition rate compared by overtime.
PercentSalaryHike	Numerical	Included in numerical summary/correlation analysis.
PerformanceRating	Numerical	Included in numerical summary/correlation analysis.
RelationshipSatisfaction	Numerical	Included in numerical summary/correlation analysis.
StandardHours	Numerical	Constant feature; removed.
StockOptionLevel	Numerical	Included in numerical summary/correlation analysis.
TotalWorkingYears	Numerical	Included in numerical summary/correlation analysis.
TrainingTimesLastYear	Numerical	Included in numerical summary/correlation analysis.
WorkLifeBalance	Numerical	Included in numerical summary/correlation analysis.
YearsAtCompany	Numerical	Compared with income and included in outlier analysis.
YearsInCurrentRole	Numerical	Included in numerical summary/correlation analysis.
YearsSinceLastPromotion	Numerical	Included in numerical summary/correlation analysis.
YearsWithCurrManager	Numerical	Included in numerical summary/correlation analysis.

3. DATA UNDERSTANDING

3.1 Dataset Dimensions

The initial DataFrame shape is (1470, 35), meaning that the notebook begins with 1,470 employee records and 35 columns. After removing EmployeeCount, Over18, StandardHours, and EmployeeNumber, the resulting DataFrame shape is (1470, 31).

3.2 Data Types

Data type	Count	Features
Numerical / int64	26	Age, DailyRate, DistanceFromHome, Education, EmployeeCount, EmployeeNumber,

Data type	Count	Features
		EnvironmentSatisfaction, HourlyRate, JobInvolvement, JobLevel, JobSatisfaction, MonthlyIncome, MonthlyRate, NumCompaniesWorked, PercentSalaryHike, PerformanceRating, RelationshipSatisfaction, StandardHours, StockOptionLevel, TotalWorkingYears, TrainingTimesLastYear, WorkLifeBalance, YearsAtCompany, YearsInCurrentRole, YearsSinceLastPromotion, YearsWithCurrManager
Categorical / str	9	Attrition, BusinessTravel, Department, EducationField, Gender, JobRole, MaritalStatus, Over18, OverTime

3.3 Numerical Descriptive Statistics

The notebook uses df.describe() to summarize 26 numerical features. Selected descriptive statistics that are especially relevant to the subsequent EDA are shown below.

Note: Values marked with an em dash are not reproduced here because the visible notebook output used for this report did not expose those particular mean/standard-deviation values in a complete, independently readable form. The report does not infer them.

Feature	Mean	Std. Dev.	Min	Median	Max
Age	36.924	9.135	18	36	60
DailyRate	802.486	403.509	102	802	1499
DistanceFromHome	9.193	8.107	1	7	29
Education	2.913	1.024	1	3	5
EnvironmentSatisfaction	2.722	1.093	1	3	4
JobLevel	2.064	1.107	1	2	5
JobSatisfaction	—	—	1	—	4
MonthlyIncome	—	—	1009	—	19999
NumCompaniesWorked	—	—	0	—	9
TotalWorkingYears	11.280	7.780	0	10	40
WorkLifeBalance	2.761	0.706	1	3	4
YearsAtCompany	7.008	6.126	0	5	40
YearsInCurrentRole	4.229	3.623	0	3	18
YearsSinceLastPromotion	2.188	3.222	0	1	15
YearsWithCurrManager	4.123	3.568	0	3	17

3.4 Categorical Feature Summary

Feature	Unique values	Most frequent category	Frequency
Attrition	2	No	1,233
BusinessTravel	3	Travel_Rarely	1,043
Department	3	Research & Development	961
EducationField	6	Life Sciences	606
Gender	2	Male	882
JobRole	9	Sales Executive	326
MaritalStatus	3	Married	673
Over18	1	Y	1,470
OverTime	2	No	1,054

3.5 Unique Value Analysis

Unique-value counts were examined for every feature. Three fields—EmployeeCount, Over18, and StandardHours—contain exactly one unique value. The notebook explicitly identifies these as features without useful variation and removes them before subsequent analysis. EmployeeNumber has 1,470 unique values and is treated as an identifier rather than a meaningful analytical feature.

3.6 Initial Observations

- The dataset is structurally complete in terms of non-null values: every column contains 1,470 non-null records.
- The target variable is imbalanced, with substantially more No than Yes attrition outcomes.
- The categorical summary shows three departments, nine job roles, and two overtime categories.
- Several numerical variables have meaningful variation and are subsequently used in distribution, relationship, correlation, or outlier analysis.
- EmployeeCount, Over18, and StandardHours are constant and therefore removed; EmployeeNumber is removed as an identifier.

4. DATA QUALITY AND PREPROCESSING

4.1 Missing Value Analysis

The notebook evaluates df.isnull().sum() for all 35 initial features. Every feature has 0 missing values. Therefore, no missing-value imputation or deletion is performed.

Check	Result	Decision
Missing values across all initial features	0	No missing-value treatment required

4.2 Duplicate / Data Quality Checks

An explicit duplicate-row check is not present in the supplied notebook. Consequently, this report does not claim that duplicate rows were tested or removed. This is a documented gap relative to a general EDA checklist and should be addressed if the project is extended.

4.3 Invalid or Constant Features

The notebook explicitly checks unique values and identifies EmployeeCount, Over18, and StandardHours as constant features. Their lack of variation makes them uninformative for comparative analysis. EmployeeNumber is also removed because it functions as an identifier rather than a meaningful analytical feature.

4.4 Feature Removal

Removed feature	Reason recorded in notebook
EmployeeCount	Only one unique value; no useful variation.
Over18	Only one unique value; no useful variation.
StandardHours	Only one unique value; no useful variation.

Removed feature	Reason recorded in notebook
EmployeeNumber	Identifier rather than a meaningful predictive feature.

4.5 Outlier Analysis

The notebook performs visual outlier analysis after feature removal using boxplots for Age, MonthlyIncome, DistanceFromHome, and YearsAtCompany. These plots identify observations that fall unusually far from the central distribution. The notebook does not label such observations as erroneous and does not remove them.

4.6 Preprocessing Decisions

The preprocessing performed in the supplied EDA notebook is intentionally limited: the dataset is loaded, inspected, checked for missing values, examined for unique-value structure, and reduced by removing three constant features and one identifier. The project does not document imputation, scaling, normalization, categorical encoding, or outlier deletion within this EDA notebook.

4.7 Justification for Cleaning Decisions

The cleaning decisions are directly supported by the notebook: there are no missing values requiring treatment, constant fields contribute no variation, and EmployeeNumber is an identifier. Retaining potential outliers preserves the observed workforce data rather than assuming that unusual values are errors.

5. EXPLORATORY DATA ANALYSIS

The EDA is organized around univariate distributions and bivariate/multivariate relationships. The original notebook figures are reproduced below without recreating or altering their analytical content.

5.1 Target Variable Analysis

Attrition is the central outcome of the project. The notebook records 1,233 No and 237 Yes outcomes. This imbalance is important when interpreting counts because the larger No group dominates the overall dataset.

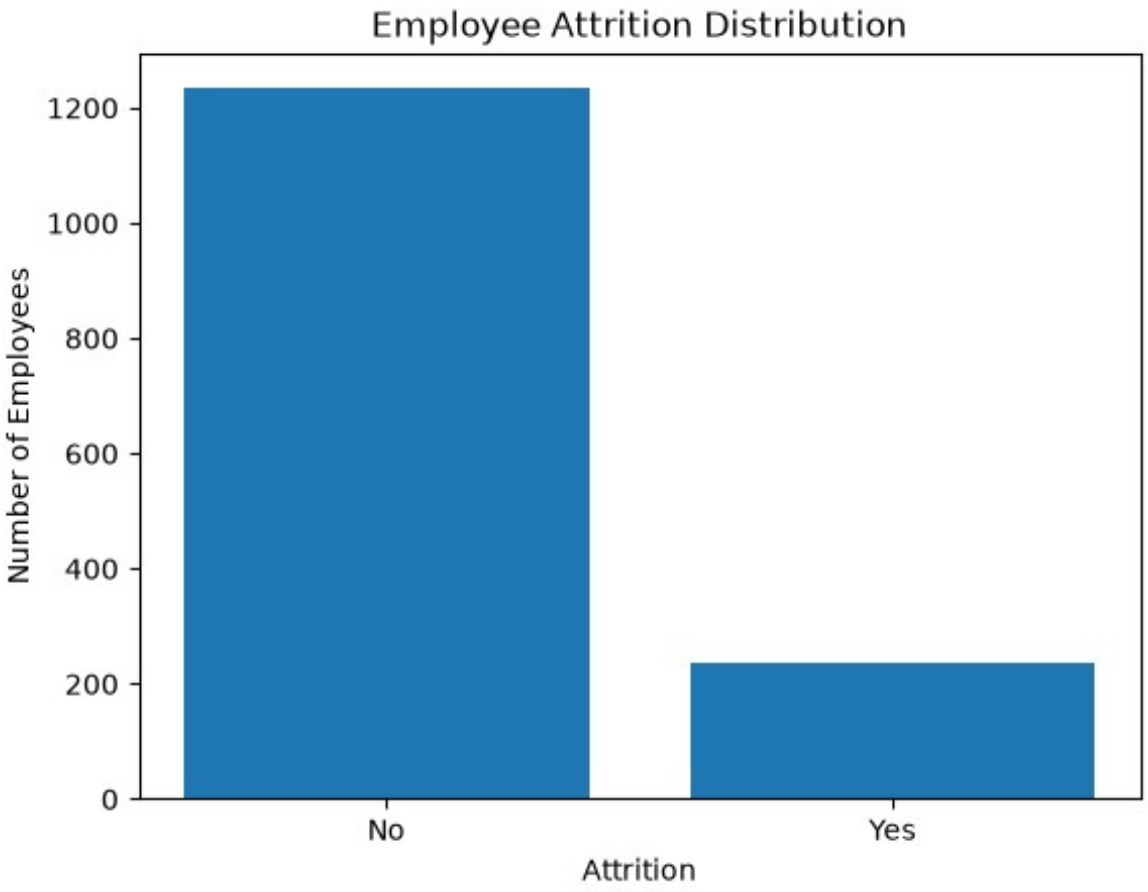


Figure 5.1. Employee Attrition Distribution.

Interpretation: The chart shows a substantially larger number of employees with Attrition = No than Attrition = Yes. The imbalance is also quantified in the notebook as 83.88% No and 16.12% Yes.

5.2 Demographic Analysis

Age and gender are examined to describe workforce composition. Age is treated as a numerical variable and visualized with a histogram, while Gender is summarized through a bar chart.

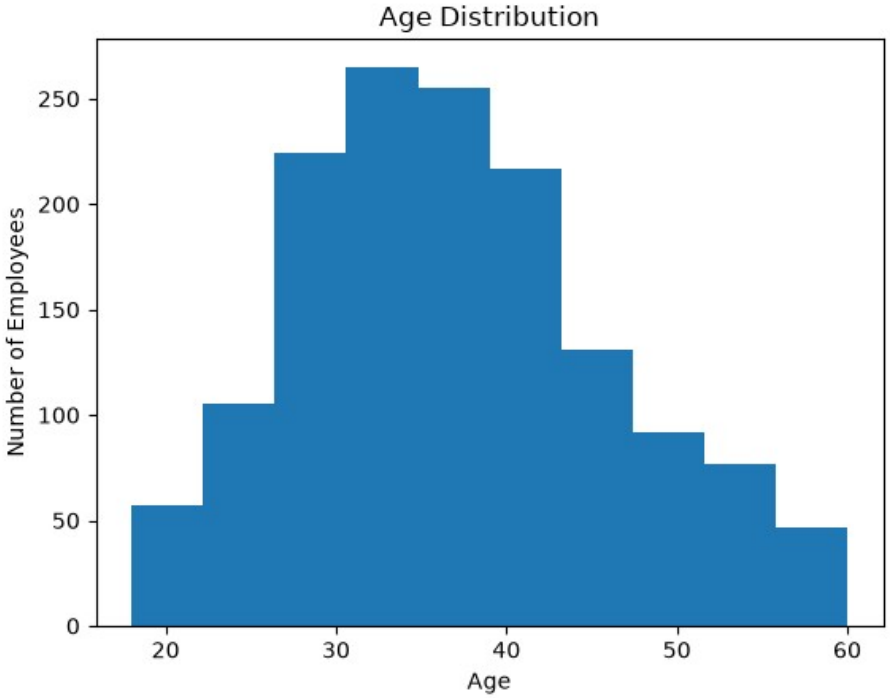


Figure 5.2. Age Distribution.

Interpretation: The histogram shows the distribution of employee ages across the dataset. The notebook uses this as a univariate view of workforce age composition.

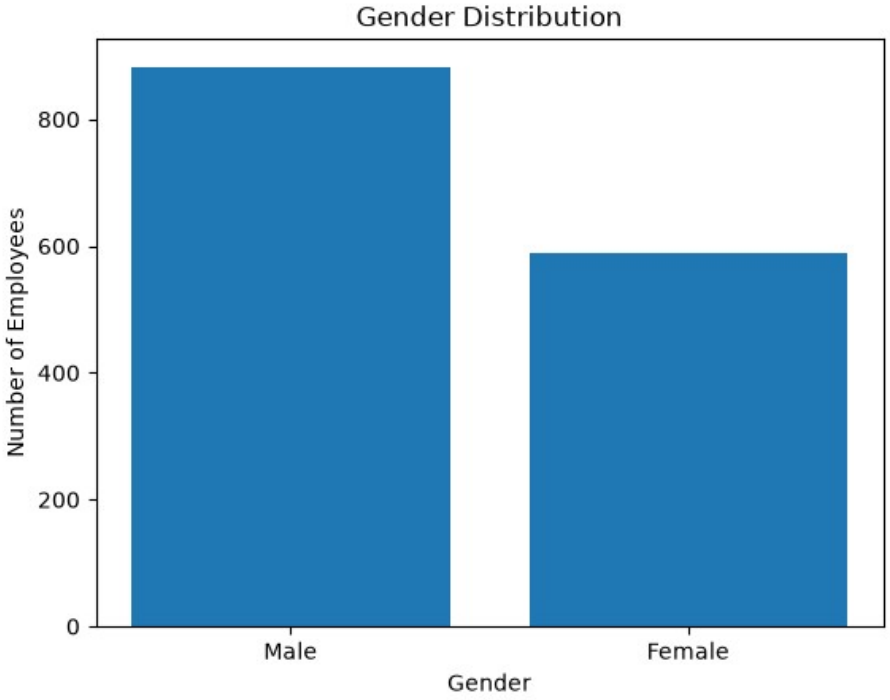


Figure 5.3. Gender Distribution.

Interpretation: The bar chart shows the number of employees in each recorded gender category. It provides a descriptive view of workforce composition.

5.3 Department Analysis

Department is examined both as a workforce-composition variable and as an attrition grouping variable. The categorical summary identifies Research & Development as the most frequent department, with 961 employees.

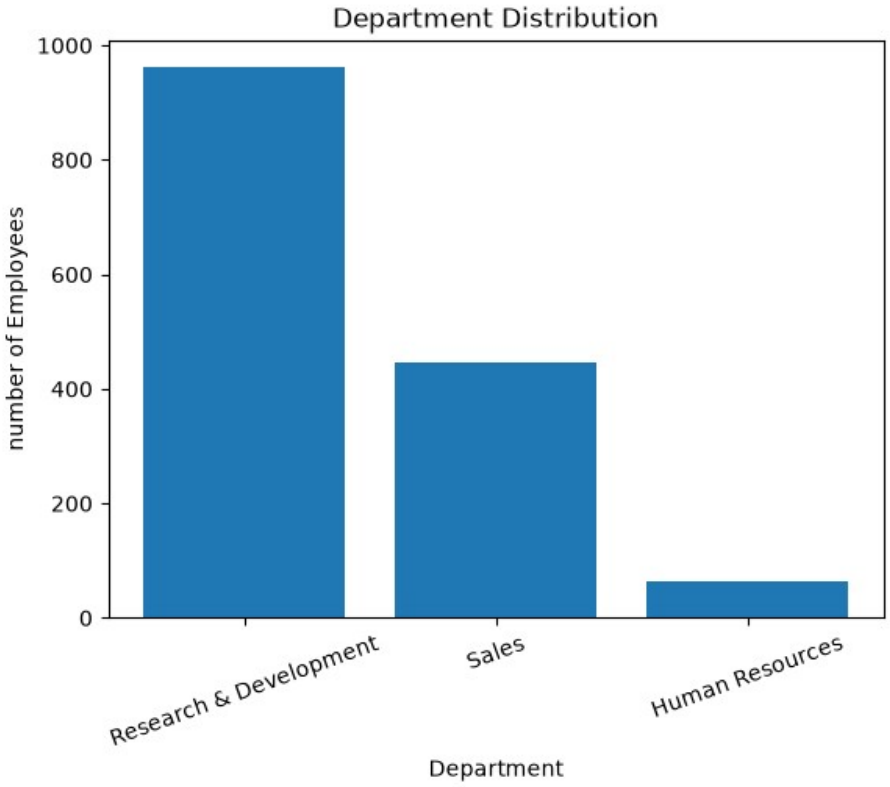


Figure 5.4. Department Distribution.

Interpretation: The chart compares the number of employees across the three departments. Research & Development is the most frequent department in the categorical summary.

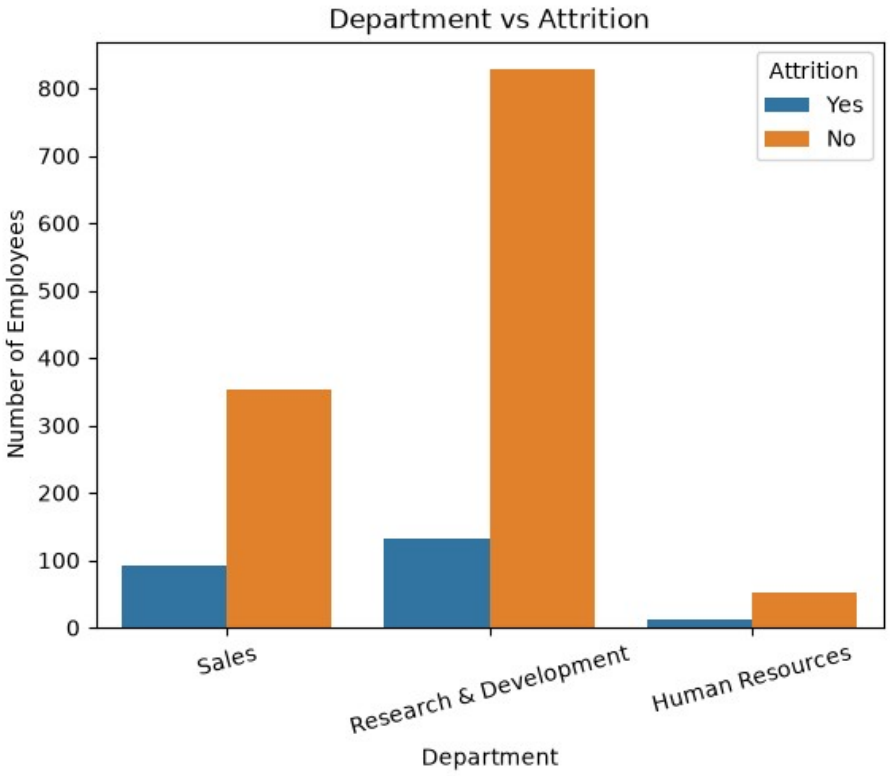


Figure 5.8. Department vs Attrition.

Interpretation: The chart compares attrition counts within departments, allowing differences in observed attrition patterns among organizational units to be examined.

5.4 Overtime Analysis

OverTime is analyzed in two stages: its overall distribution and its relationship with Attrition. The notebook explicitly describes overtime as a workplace factor that may be associated with attrition.

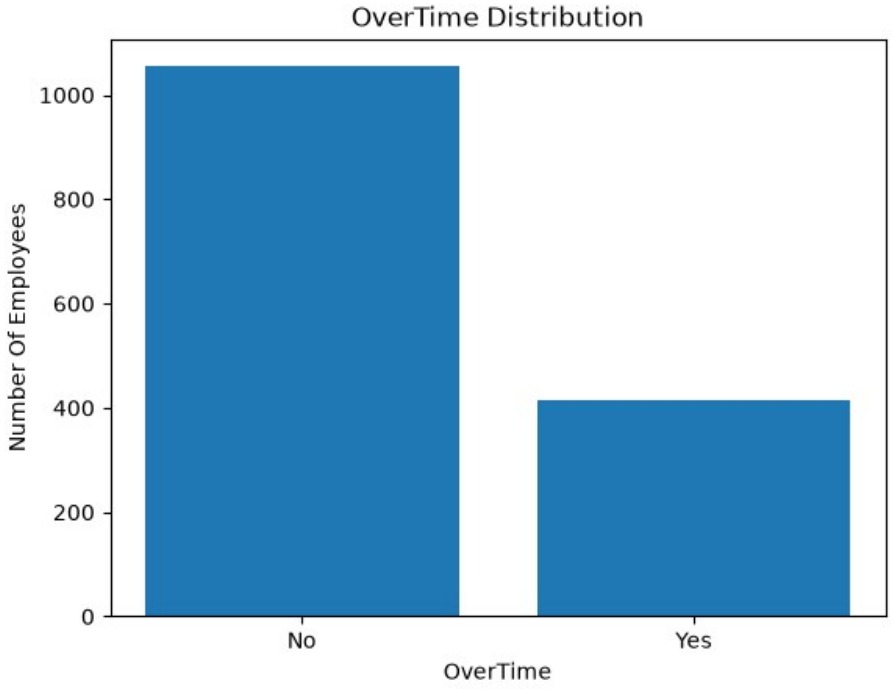


Figure 5.5. OverTime Distribution.

Interpretation: The visualization shows the distribution of overtime categories. The categorical summary records 1,054 employees with OverTime = No.

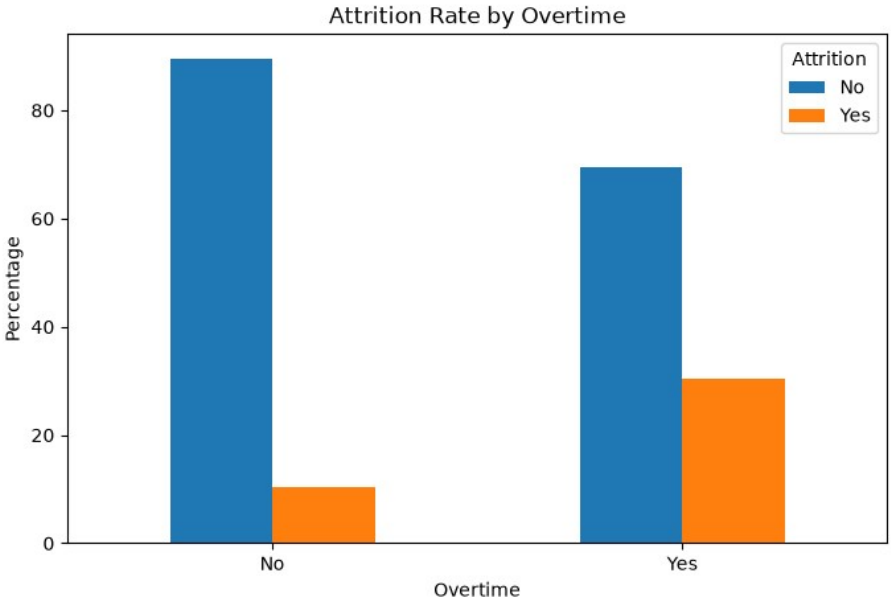


Figure 5.6. Attrition Rate by Overtime.

Interpretation: The grouped percentage chart compares Attrition outcomes within each overtime category. It provides a descriptive association between overtime status and attrition.

5.5 Job Satisfaction Analysis

Job satisfaction is compared across attrition outcomes using a countplot. The plot supports descriptive comparison of the frequency of satisfaction levels within employees who stayed and employees who left.

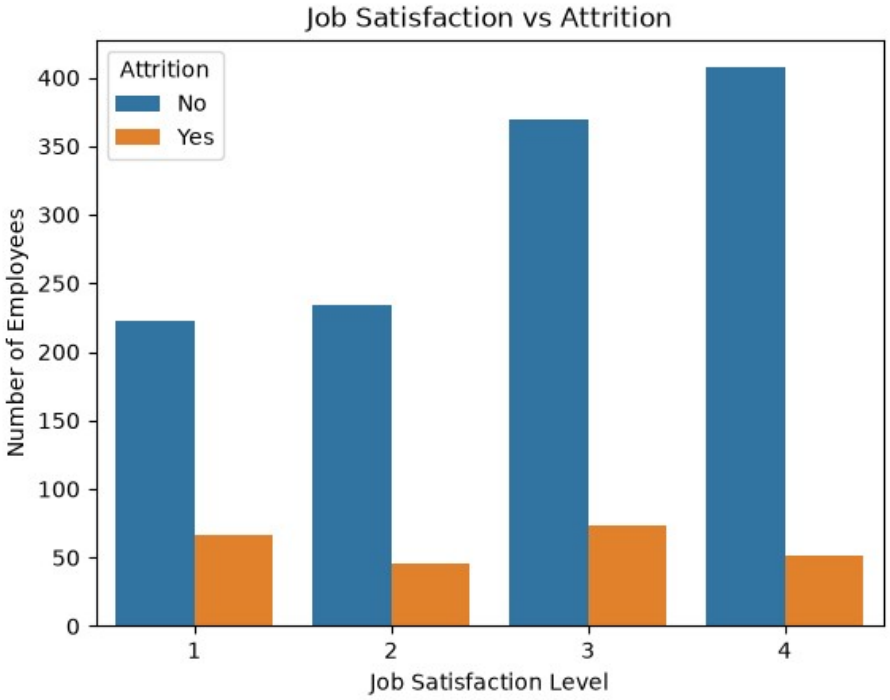


Figure 5.7. Job Satisfaction vs Attrition.

Interpretation: The countplot compares the number of employees across job-satisfaction levels for the two attrition outcomes. It is used to examine whether the distribution of satisfaction levels differs between employees who stayed and those who left.

5.6 Age Analysis

Age is analyzed both as a general workforce distribution and by attrition outcome. The age boxplot enables comparison of central tendency and spread between the two attrition groups.

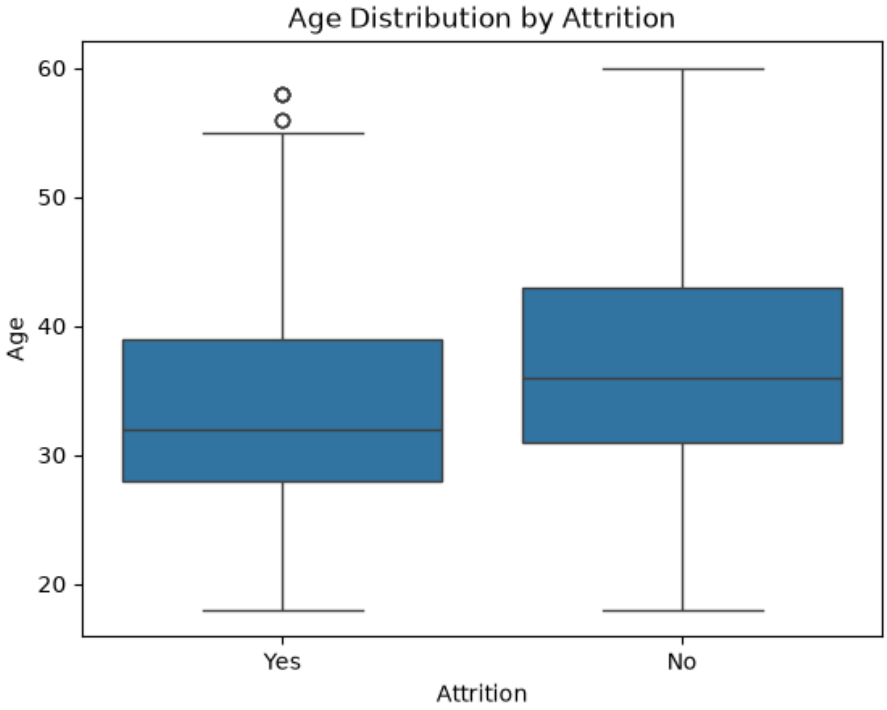


Figure 5.9. Age Distribution by Attrition.

Interpretation: The boxplot compares age distributions for employees who stayed and those who left. Differences in medians, spread, and potential extreme observations can be inspected descriptively.

5.7 Income Analysis

MonthlyIncome is examined by attrition outcome and in relation to Age and YearsAtCompany. The analysis is descriptive and is not interpreted as evidence that income causes attrition.

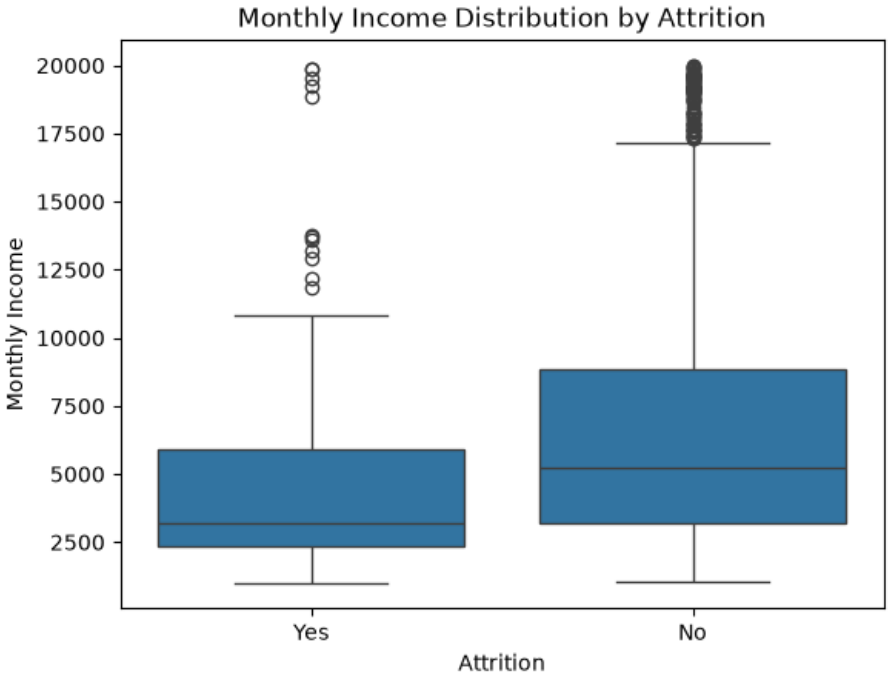


Figure 5.10. Monthly Income Distribution by Attrition.

Interpretation: The boxplot compares monthly-income distributions by attrition outcome. It provides a descriptive view of compensation differences between the two groups.

5.8 Distance From Home Analysis

DistanceFromHome is compared between attrition groups with a boxplot. The purpose is to inspect whether the distribution differs descriptively between employees who stayed and those who left.

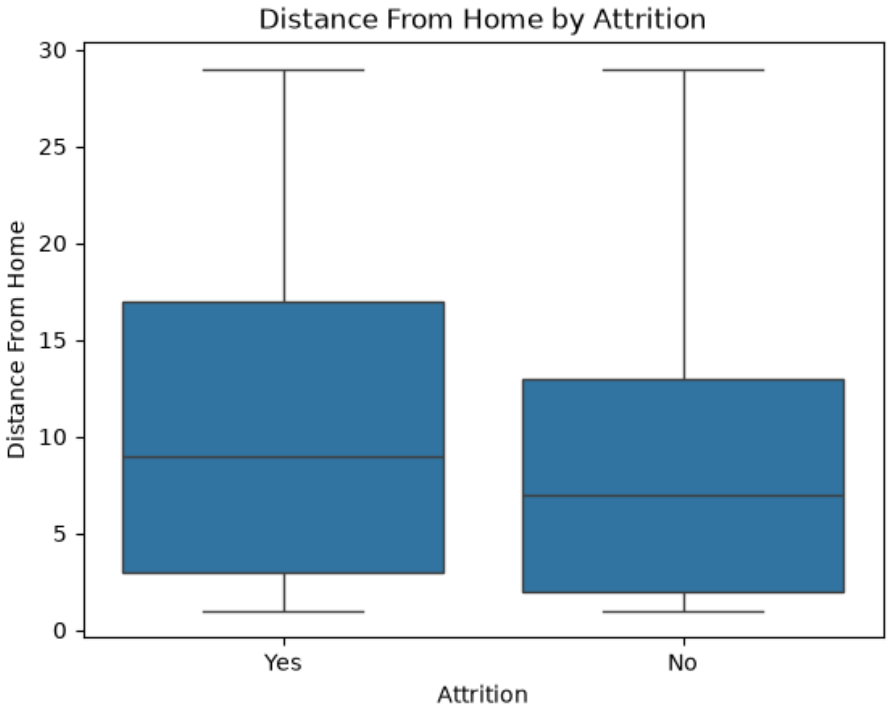


Figure 5.11. Distance From Home by Attrition.

Interpretation: The boxplot compares distance-from-home distributions for the two attrition outcomes. It is used as a descriptive indicator of how commuting distance varies between groups.

5.9 Correlation Analysis

The correlation heatmap summarizes pairwise relationships among numerical variables. The most visible positive relationships include JobLevel with MonthlyIncome (approximately 0.95), JobLevel with TotalWorkingYears (approximately 0.78), MonthlyIncome with TotalWorkingYears (approximately 0.77), and YearsAtCompany with YearsWithCurrManager (approximately 0.77). These are descriptive linear associations, not causal findings.

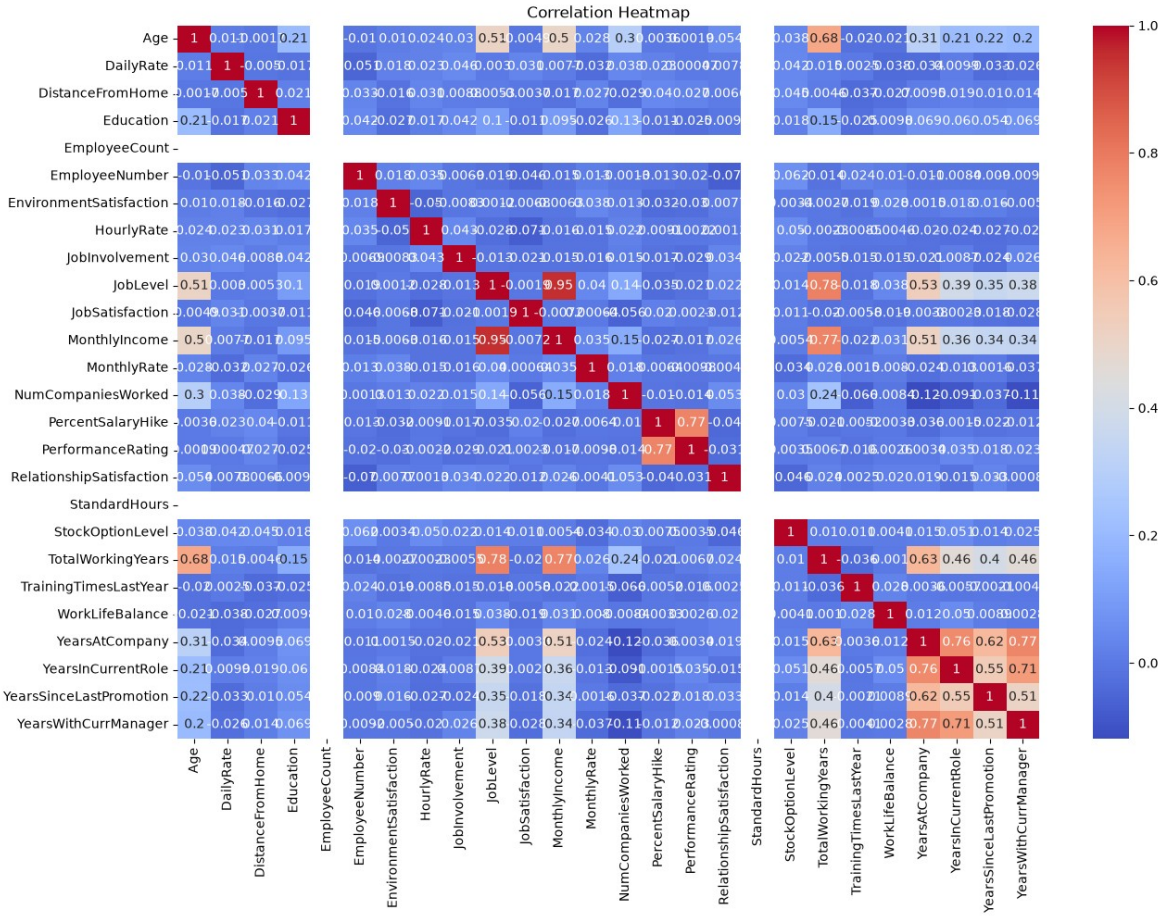


Figure 5.14. Correlation Heatmap.

Interpretation: The heatmap presents pairwise Pearson-style linear correlations among the numerical variables included by df.corr(numeric_only=True). Strong positive relationships are visibly concentrated among job level, income, total working years, and tenure-related variables.

5.10 Relationship Analysis

The two scatterplots provide multivariate visual views by combining two numerical variables with Attrition as a grouping variable. They help inspect the distribution of employee groups rather than establish predictive or causal relationships.

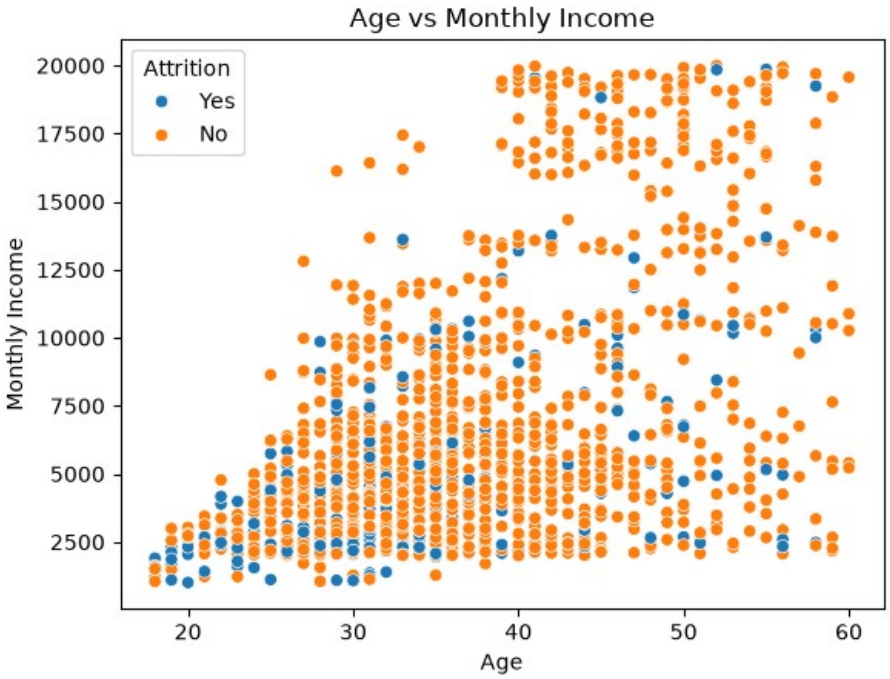


Figure 5.12. Age vs Monthly Income.

Interpretation: The scatterplot examines the relationship between age and monthly income while using attrition as a grouping variable. It provides a multivariable view of how the two numerical variables are distributed across attrition categories.

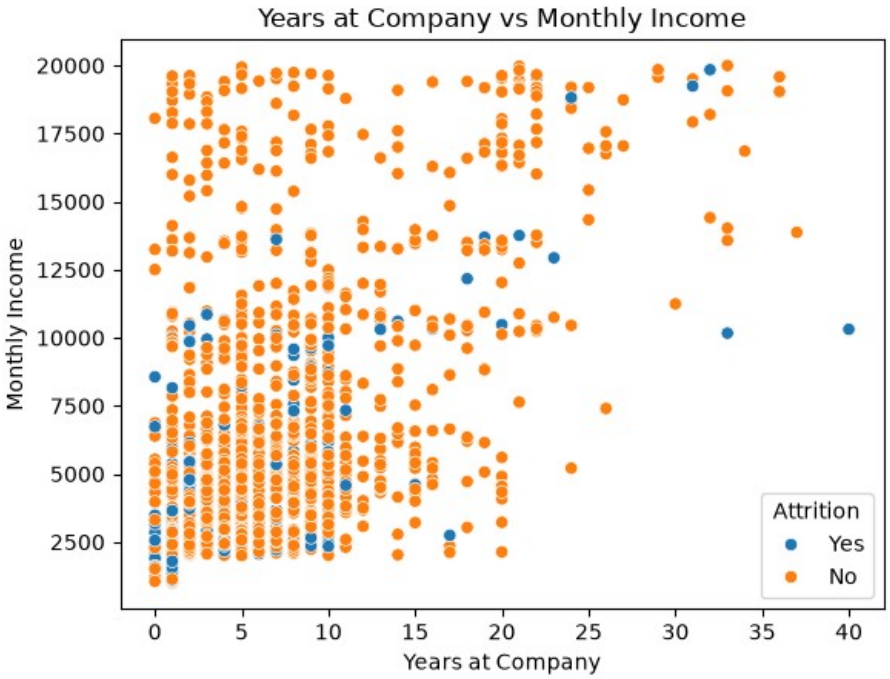


Figure 5.13. Years at Company vs Monthly Income.

Interpretation: The scatterplot examines the relationship between tenure at the company and monthly income, with attrition shown as a grouping variable.

6. KEY FINDINGS AND RISK INDICATORS

The following findings are limited to patterns visible in the supplied notebook. They are presented as observations or associations rather than causal explanations.

Attrition imbalance: Attrition is unevenly distributed: 83.88% of records are No and 16.12% are Yes. This means simple employee counts should be interpreted with awareness of the class imbalance.

Overtime association: The overtime analysis shows a noticeable descriptive relationship between overtime status and attrition. The chart compares attrition percentages within overtime categories.

Job satisfaction: The job-satisfaction countplot shows different frequencies of satisfaction levels across the two attrition outcomes, indicating a descriptive association worth further investigation.

Department differences: The department-versus-attrition plot shows that attrition patterns vary across departments. This is a group-level descriptive observation.

Age and income: The age and monthly-income boxplots show differences in the distributions of these variables between attrition groups.

Distance from home: Distance from home provides another descriptive comparison between employees who stayed and those who left.

Tenure and income: The YearsAtCompany versus MonthlyIncome scatterplot indicates a relationship between tenure and income, while attrition is shown as a grouping variable.

Numerical correlations: The heatmap reveals strong positive correlations among several career-level, income, and tenure variables, including JobLevel–MonthlyIncome and YearsAtCompany–YearsWithCurrManager.

6.1 Risk Indicators: Appropriate Interpretation

Within an EDA context, the variables above can be treated as candidate risk indicators or investigation areas because their distributions or associations differ across attrition outcomes. The notebook does not establish that any of these factors independently causes attrition. Any operational HR decision would therefore require additional statistical validation and contextual analysis.

7. STATISTICAL / CORRELATION ANALYSIS

The notebook performs correlation analysis using df.corr(numeric_only=True), producing a correlation matrix for numerical variables. The resulting heatmap displays both direction and magnitude of pairwise linear association.

7.1 Interpretation of Important Correlations

Variable pair	Approx. correlation visible	Interpretation
JobLevel – MonthlyIncome	~0.95	Very strong positive linear association visible in the heatmap.
JobLevel – TotalWorkingYears	~0.78	Strong positive association.
MonthlyIncome – TotalWorkingYears	~0.77	Strong positive association.
YearsAtCompany – YearsWithCurrManager	~0.77	Strong positive association.
YearsAtCompany – YearsInCurrentRole	~0.76	Strong positive association.
Age – TotalWorkingYears	~0.68	Moderately strong positive association.

Variable pair	Approx. correlation visible	Interpretation
YearsAtCompany – YearsSinceLastPromotion	~0.62	Moderate-to-strong positive association.

These values are read from the notebook's displayed heatmap and are rounded for reporting. The correlations indicate linear association between numerical variables; they do not by themselves identify causal mechanisms or prove that a variable is a determinant of attrition.

7.2 Correlation and Attrition

Attrition is categorical and is not itself part of the numerical correlation heatmap produced by df.corr(numeric_only=True). Attrition relationships are therefore examined separately through counts, grouped percentages, boxplots, and scatterplots.

8. OUTLIER ANALYSIS

Outlier analysis is performed with boxplots for four numerical variables: Age, MonthlyIncome, DistanceFromHome, and YearsAtCompany. The purpose is to identify observations that fall unusually far from the central range of each variable.

8.1 Age

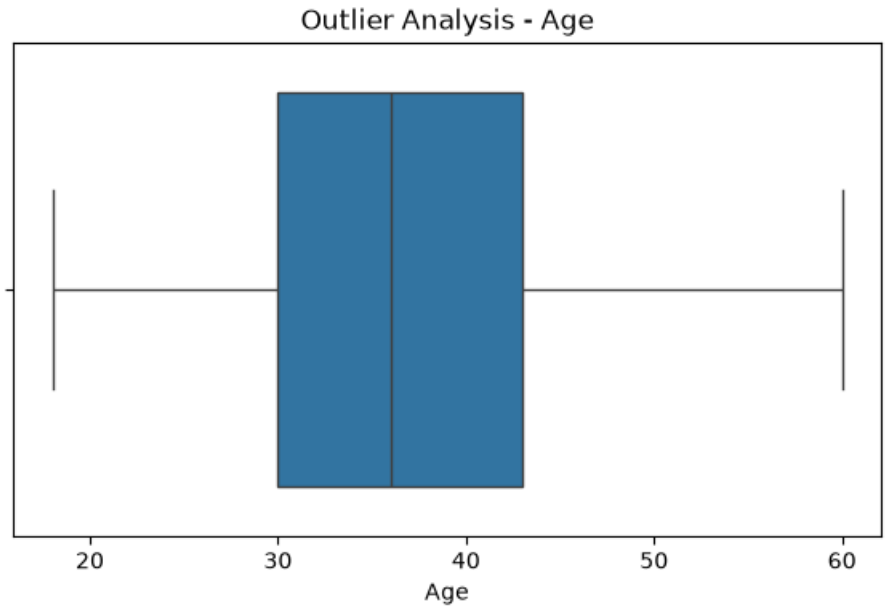


Figure 5.15. Outlier Analysis – Age.

Interpretation: The boxplot provides a univariate view of potential unusually low or high age observations.

8.2 Monthly Income

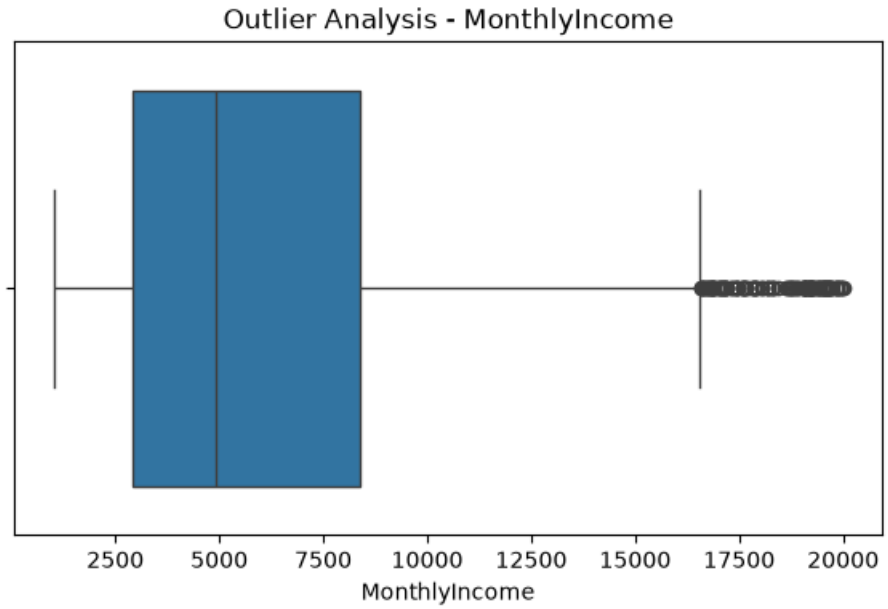


Figure 5.16. Outlier Analysis – MonthlyIncome.

Interpretation: The boxplot highlights the spread of monthly income and potential observations far from the central range.

8.3 Distance From Home

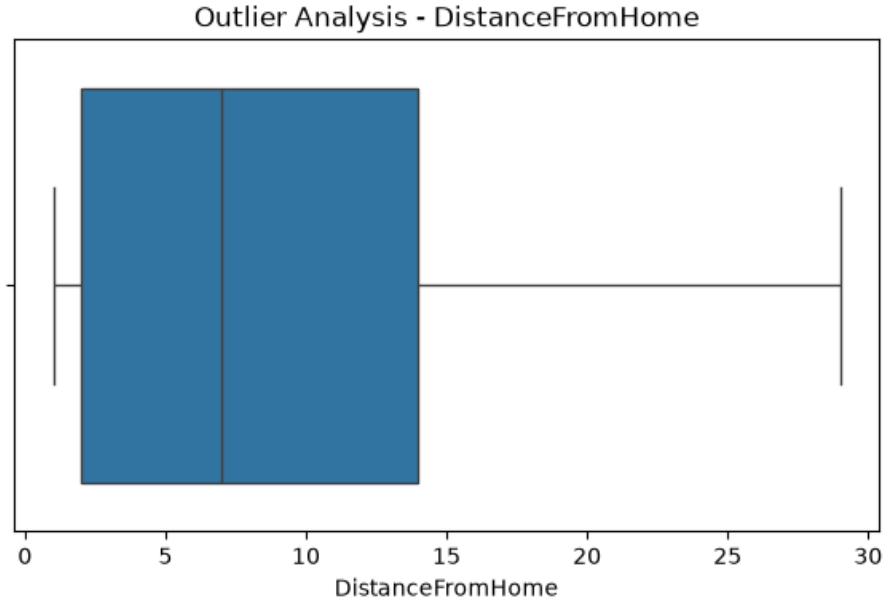


Figure 5.17. Outlier Analysis – DistanceFromHome.

Interpretation: The boxplot shows the distribution of commuting distance and potential extreme observations.

8.4 Years at Company

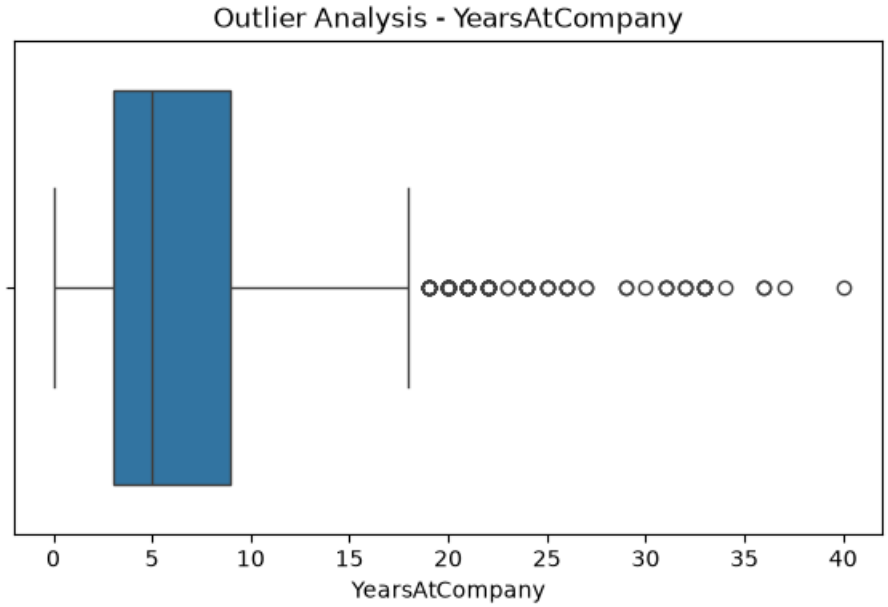


Figure 5.18. Outlier Analysis – YearsAtCompany.

Interpretation: The boxplot shows the tenure distribution and potential observations outside the central range.

The notebook does not remove the observations identified visually as potential outliers. In a workforce dataset, unusual values may represent legitimate employee situations, so treating them as data errors would require additional evidence.

9. AI & GENAI USAGE

AI/GenAI assistance was used as a supporting tool for organizing the academic report, improving explanation clarity, and structuring the documentation around the completed EDA workflow. The actual dataset inspection, calculations, visualizations, and observations reported here are grounded in the supplied project notebook.

No AI-generated statistic, graph, model result, or observation has been presented as project evidence unless it is supported by the notebook. The report therefore distinguishes documentation assistance from analytical execution.

The supplied notebook itself does not contain a formal log of GenAI prompts or AI-generated analytical outputs. This section consequently does not claim that GenAI performed the EDA.

10. PROJECT WORKFLOW

The project follows the workflow below, reconstructed from the sequence of operations documented in the notebook.

Stage	Component	Activity
1	Dataset	Load the employee attrition CSV into a

Stage	Component	Activity
		Pandas DataFrame.
2	Data Understanding	Inspect dimensions, structure, data types, numerical and categorical summaries.
3	Data Quality Checks	Check missing values and unique-value structure.
4	Exploratory Data Analysis	Analyze target and selected workforce characteristics.
5	Visualization	Use bar charts, histogram, countplots, boxplots, and scatterplots.
6	Correlation Analysis	Compute and visualize numerical correlations.
7	Outlier Analysis	Inspect Age, MonthlyIncome, DistanceFromHome, and YearsAtCompany using boxplots.
8	Key Findings	Summarize descriptive patterns and relationships.
9	Conclusion	State what the EDA demonstrates and identify reasonable next steps.

10.1 Workflow Diagram

Dataset

↓

Data Understanding

↓

Data Quality Checks

↓

Exploratory Data Analysis

↓

Visualization

↓

Correlation Analysis

↓

Outlier Analysis

↓

Key Findings

↓

Conclusion

11. CONCLUSION

The Employee Attrition & Workforce Analytics project provides a structured exploratory view of an employee workforce dataset containing 1,470 records and 35 initial features. The analysis establishes the dataset structure, summarizes numerical and categorical variables, verifies that no missing values are present, identifies constant features, and removes constant fields together with the employee identifier.

The EDA shows that attrition is unevenly distributed, with 237 employees recorded as leaving and 1,233 recorded as not leaving. The notebook then examines descriptive relationships involving overtime, job satisfaction, department, age, monthly income, distance from home, and tenure. The correlation heatmap additionally reveals strong numerical relationships among job level, income, total working years, and tenure-related variables.

Overall, the project demonstrates how EDA can be used to move from raw workforce data toward interpretable patterns and candidate areas for further investigation. The findings remain descriptive: the analysis does not establish that any individual factor causes employee attrition.

12. LIMITATIONS

- The analysis is based on one employee dataset and therefore describes the patterns represented in that dataset.
- The notebook is observational and exploratory; EDA alone cannot establish causation.
- The target variable is imbalanced, with the No category substantially larger than the Yes category.
- An explicit duplicate-row check is not documented in the supplied notebook.
- Explicit validity/range checks are not documented as a separate preprocessing stage in the supplied notebook.
- The notebook contains one histogram rather than a larger set of histograms that may be specified in a generic EDA checklist.
- The analysis is primarily descriptive and does not include formal hypothesis testing or predictive-model evaluation.
- The dataset source is identified in the notebook by its CSV filename, but no external publisher or URL is documented there.

13. FUTURE SCOPE

The completed EDA provides a foundation for extending the project beyond descriptive analysis. Reasonable next steps, consistent with the project scope, include:

- Perform an explicit duplicate-row and validity/range check to complete the data-quality checklist.
- Conduct deeper statistical tests for selected relationships identified during EDA.
- Investigate the imbalance in the Attrition target using appropriate analytical techniques.
- Develop engineered features where justified by domain context.
- Apply predictive classification models to estimate attrition risk and evaluate them with metrics appropriate for an imbalanced target.
- Compare models and validate results using suitable train/test or cross-validation procedures.

- Use explainable-AI techniques to understand and communicate model-level feature contributions.
- Validate any HR-oriented risk indicators against organizational context before operational use.

14. REFERENCES

1. Employee Attrition & Workforce Analytics project notebook, “01_Data_Understanding(8).ipynb”, supplied as the source of truth for the analysis, figures, observations, and results.

2. BLEEP Education, Artificial Intelligence Internship – EDA/project reference material and EDA report template, available in the project materials.

3. Dataset file referenced by the notebook: “WA_Fn-UseC_-HR-Employee-Attrition.csv”. The notebook does not document an external publisher or URL; therefore no external attribution is added here.

15. APPENDIX

Appendix A. BLEEP EDA Requirement Alignment

BLEEP requirement	Status	Evidence / note
Data understanding	Covered	Dataset loaded and structurally inspected.
Dataset dimensions	Covered	1470 × 35 initially; 1470 × 31 after removal.
Data types	Covered	26 numerical and 9 categorical/string features.
Descriptive statistics	Covered	df.describe() and categorical summary.
Data dictionary	Partially covered	Feature inventory is documented; full semantic descriptions are not present in the notebook.
Missing-value analysis	Covered	All 35 initial columns have 0 missing values.
Duplicate checks	Not documented	No explicit duplicate-row check appears in the supplied notebook.
Data validity checks	Not explicitly documented	Unique-value analysis is present, but a separate validity/range-check stage is not documented.
Outlier analysis	Covered	Four numerical variables are inspected with boxplots.
Univariate analysis	Covered	Target, age, gender, department, and overtime distributions are visualized.
Bivariate analysis	Covered	Overtime, job satisfaction, department, age, income, and distance relationships are examined.
Multivariate analysis	Covered	Correlation heatmap and grouped scatterplots provide multivariable views.
3 Histograms	Partially covered	One histogram is present: Age Distribution.
3 Box plots	Covered	Three attrition-comparison boxplots plus four outlier-analysis boxplots are present.
3 Bar charts	Covered	Multiple bar charts are present for attrition, gender, department, and overtime.
2 Count plots	Covered	Job Satisfaction vs Attrition and Department vs Attrition are present, plus the attrition-related count comparison.
2 Scatter plots	Covered	Age vs MonthlyIncome and YearsAtCompany vs MonthlyIncome.
Correlation heatmap	Covered	Numerical correlation heatmap included.
2 additional meaningful visualizations	Covered through additional relationship plots	Overtime attrition-rate and multiple group-comparison plots are included.
Interpretation of visualizations	Covered	Each major figure is followed by an interpretation in this report.
Key findings	Covered	Section 6.
Conclusion and next steps	Covered	Sections 11 and 13.

Appendix B. Supplementary Project Notes

The notebook uses Pandas, NumPy, Matplotlib, and Seaborn for the documented EDA workflow. The original notebook figures reproduced in this report were extracted directly from the notebook outputs; no replacement charts were created.